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FOR IBPS SBI PO/CLERK PRELIMS 2026

QUANT CHECKLIST

the PRACTICE PAPER

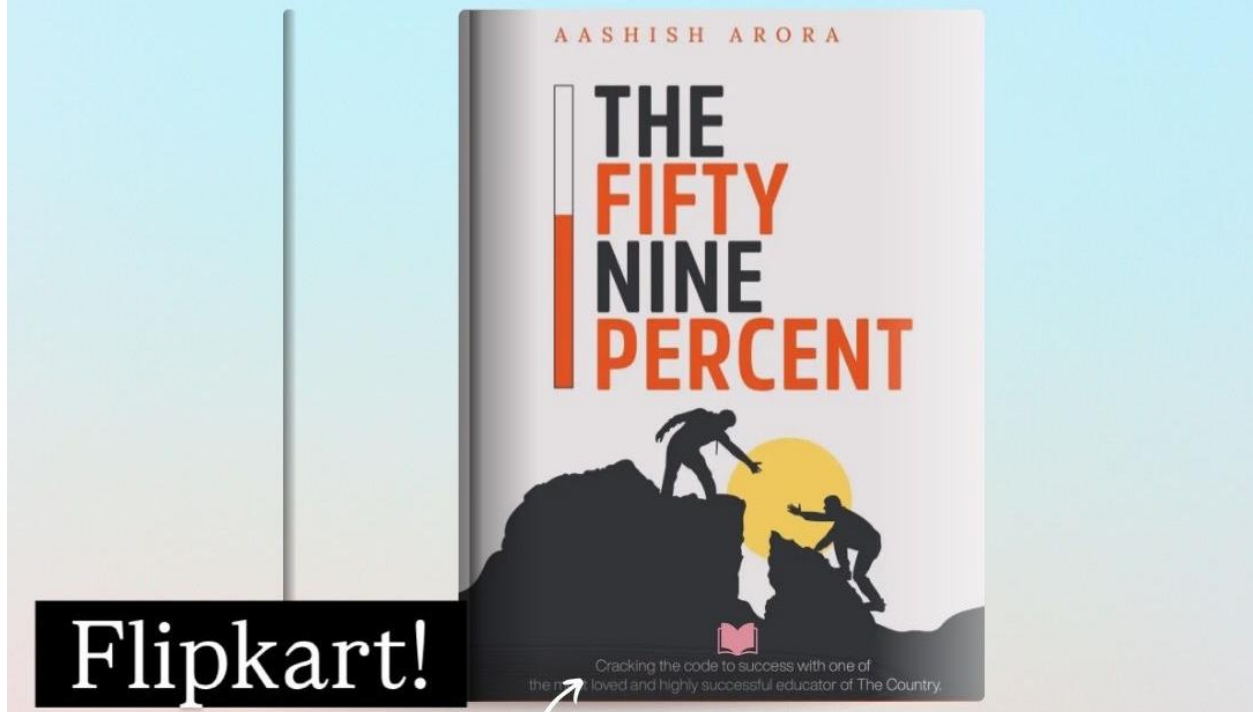


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Dear Students,

This daily practice sheet is designed to build both speed and accuracy, one day at a time.

It contains a mix of easy, moderate, and challenging questions to prepare you for every possible scenario in the exam. Treat it like a warm-up before the real game.

Solve it daily without fail. Don't wait for motivation—show up with discipline. Because it's not talent but consistent hard work that takes you places.

Stay focused. Stay consistent. Work hard.

-Aashish Arora

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1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. $37.5\% \text{ of } 864 + \sqrt{4096} - \frac{5}{11} \text{ of } 440 = ? \times 8$

- A. 23.5
- B. 21.5
- C. 25.5
- D. 27.5
- E. None of these

2. $? \% \text{ of } 900 + \frac{7}{12} \text{ of } 480 = 62.5\% \text{ of } 640 + 420$

- A. 56
- B. 60
- C. 64
- D. 68
- E. None of these

3. $(\sqrt{9801} - \sqrt[3]{91125}) \times 6 + 68.75\% \text{ of } 512 = ? + 231$

- A. 425
- B. 435
- C. 445

D. 455

- E. None of these

4. $[(15^2 - 9^2) \div 4] + 58.33\% \text{ of } 720 - \frac{1}{13} \text{ of } 780 = ? \times 5$

- A. 79.2
- B. 78.4
- C. 80.6
- D. 81.4
- E. None of these

5. $6\frac{2}{3} + 5\frac{1}{3} - 3\frac{3}{4} + 1\frac{3}{4} = ?$

- A. 8
- B. 9
- C. 11
- D. 10
- E. None of these

6. $14.28\% \text{ of } 1680 + \frac{5}{6} \text{ of } 504 - \sqrt{2916} = ? + 337$

- A. 259
- B. 269
- C. 279

D. 289

E. None of these

7. $[27.27\% \text{ of } 1210 + 45.45\% \text{ of } 770] \div 5 + 4^3 = ?$

A. 192

B. 196

C. 204

D. 200

E. None of these

8. $\frac{11}{12} \text{ of } 360 + 16.66\% \text{ of } 540 - 4^2 = ? \times 7 + 47$

A. 49

B. 51

C. 53

D. 55

E. None of these

9. $125\% \text{ of } 336 - 31.25\% \text{ of } 768 + \sqrt[3]{46656} = ? \times 3$

A. 72

B. 68

C. 76

D. 80

E. None of these

10. $(? \div 8) + \frac{7}{15} \text{ of } 450 = 88.88\% \text{ of } 900 - 18^2$

A. 2116

B. 2144

C. 2128

D. 2088

E. None of these

11. $52\% \text{ of } 1250 + \frac{13}{40} \text{ of } 800 - 7^2 = ? + 511$

A. 350

B. 340

C. 360

D. 370

E. None of these

12. $[\frac{6}{7} \text{ of } 441 + \frac{9}{16} \text{ of } 384] - \sqrt[3]{17576} = ? + 449$

A. 111

B. 115

C. 123

D. 119

E. None of these

13. $83.33\% \text{ of } 612 + \sqrt{2704} + 2^3 \times 11 = ? \times 13$

A. 46

- B. 50
C. 54
D. 58
E. None of these

14. $\sqrt{?} + 75\% \text{ of } 360 = 16^2 + 100$

- A. 7056
B. 7225
C. 7396
D. 7569
E. None of these

15. $(\frac{3}{8} \text{ of } 640 \div 12) + \frac{5}{9} \text{ of } 216 + 6^3 \div 9 = ?$

- A. 164
B. 156
C. 160
D. 168
E. None of these

16. $? - 12.5\% \text{ of } 928 = \frac{4}{9} \text{ of } 810 + 13^2$

- A. 635
B. 645
C. 655
D. 665
E. None of these

17. $26 \times 18 - 14 \times 12 + 56.25\% \text{ of } 320 = ? \times 10$

- A. 44
B. 46
C. 48
D. 52
E. None of these

18. $\frac{7}{11} \text{ of } 704 + \frac{5}{12} \text{ of } 720 - 68.75\% \text{ of } 576 = ? + 74$

- A. 268
B. 274
C. 282
D. 278
E. None of these

19. $(\sqrt{7056} + \sqrt[3]{15625}) \times 5 - 37.5\% \text{ of } 480 = ?$

- A. 355
B. 365
C. 375
D. 385
E. None of these

20. $(72.72\% \text{ of } 605 + 32.5\% \text{ of } 800) \div ? = \sqrt{2025} + 5^2$

- A. 10
B. 8

- C. 12
D. 14
E. None of these

Answers:

1. A
2. B
3. C
4. A
5. D
6. B
7. D
8. B
9. A
10. C
11. A
12. D
13. B
14. C
15. A
16. B
17. C
18. D
19. B
20. A

Solutions:

$$1. 37.5\% \text{ of } 864 + \sqrt{4096} - \frac{5}{11} \text{ of } 440 = ? \times 8$$

$$37.5\% = \frac{3}{8} \text{ and } \sqrt{4096} = 64$$

$$\frac{3}{8} \times 864 + 64 - \frac{5}{11} \times 440 = ? \times 8$$

$$324 + 64 - 200 = ? \times 8$$

$$188 = ? \times 8$$

$$? = 23.5$$

$$2. ?\% \text{ of } 900 + \frac{7}{12} \text{ of } 480 = 62.5\% \text{ of } 640 + 420$$

$$62.5\% = \frac{5}{8} \text{ and } \frac{7}{12} \text{ of } 480 = 280$$

$$?\% \times 900 + 280 = \frac{5}{8} \times 640 + 420$$

$$?\% \times 900 + 280 = 400 + 420$$

$$?\% \times 900 = 820 - 280$$

$$? = 540 \times 100 \div 900$$

$$? = 60$$

$$3. (\sqrt{9801} - \sqrt[3]{91125}) \times 6 + 68.75\% \text{ of } 512 = ? + 231$$

$$\sqrt{9801} = 99, \sqrt[3]{91125} = 45 \text{ and } 68.75\% = \frac{11}{16}$$

$$(99 - 45) \times 6 + \frac{11}{16} \times 512 = ? + 231$$

$$54 \times 6 + 352 = ? + 231$$

$$324 + 352 = ? + 231$$

$$676 - 231 = ?$$

$$? = 445$$

$$4. [(15^2 - 9^2) \div 4] + 58.33\% \text{ of } 720 - \frac{1}{13} \text{ of } 780 = ? \times 5$$

$$58.33\% = \frac{7}{12}, 15^2 = 225 \text{ and } 9^2 = 81$$

$$(225 - 81) \div 4 + \frac{7}{12} \times 720 - \frac{1}{13} \times 780 = ? \times 5$$

$$144 \div 4 + 420 - 60 = ? \times 5$$

$$36 + 420 - 60 = ? \times 5$$

$$396 = ? \times 5$$

$$? = 79.2$$

$$5. 6\frac{2}{3} + 5\frac{1}{3} - 3\frac{3}{4} + 1\frac{3}{4} = ?$$

$$= (6 + 5 - 3 + 1) + (\frac{2}{3} + \frac{1}{3} - \frac{3}{4} + \frac{3}{4})$$

$$= 9 + 1$$

$$= 10$$

$$6. 14.28\% \text{ of } 1680 + \frac{5}{6} \text{ of } 504 - \sqrt{2916} = ? + 337$$

$$14.28\% = \frac{1}{7} \text{ and } \sqrt{2916} = 54$$

$$\frac{1}{7} \times 1680 + \frac{5}{6} \times 504 - 54 = ? + 337$$

$$240 + 420 - 54 = ? + 337$$

$$606 = ? + 337$$

$$? = 269$$

$$7. [27.27\% \text{ of } 1210 + 45.45\% \text{ of } 770] \div 5 + 4^3 = ?$$

$$27.27\% = \frac{3}{11} \text{ and } 45.45\% = \frac{5}{11}$$

$$[\frac{3}{11} \times 1210 + \frac{5}{11} \times 770] \div 5 + 64 = ?$$

$$[330 + 350] \div 5 + 64 = ?$$

$$680 \div 5 + 64 = ?$$

$$136 + 64 = ?$$

$$? = 200$$

$$8. \frac{11}{12} \text{ of } 360 + 16.66\% \text{ of } 540 - 4^2 = ? \times 7 + 47$$

$$16.66\% = \frac{1}{6} \text{ and } 4^2 = 16$$

$$\frac{11}{12} \times 360 + \frac{1}{6} \times 540 - 16 = ? \times 7 + 47$$

$$330 + 90 - 16 = ? \times 7 + 47$$

$$404 = ? \times 7 + 47$$

$$? \times 7 = 357$$

$$? = 51$$

$$9. 125\% \text{ of } 336 - 31.25\% \text{ of } 768 + \sqrt[3]{46656} = ? \times 3$$

$$125\% = \frac{5}{4}, 31.25\% = \frac{5}{16} \text{ and } \sqrt[3]{46656} = 36$$

$$\frac{5}{4} \times 336 - \frac{5}{16} \times 768 + 36 = ? \times 3$$

$$420 - 240 + 36 = ? \times 3$$

$$216 = ? \times 3$$

$$? = 72$$

$$10. (? \div 8) + \frac{7}{15} \text{ of } 450 = 88.88\% \text{ of } 900 - 18^2$$

$$88.88\% = \frac{8}{9} \text{ and } 18^2 = 324$$

$$(? \div 8) + \frac{7}{15} \times 450 = \frac{8}{9} \times 900 - 324$$

$$? \div 8 + 210 = 800 - 324$$

$$? \div 8 + 210 = 476$$

$$? \div 8 = 266$$

$$? = 2128$$

$$11. 52\% \text{ of } 1250 + \frac{13}{40} \text{ of } 800 - 7^2 = ? + 511$$

$$52\% \text{ of } 1250 = 650 \text{ and } 7^2 = 49$$

$$650 + \frac{13}{40} \times 800 - 49 = ? + 511$$

$$650 + 260 - 49 = ? + 511$$

$$861 = ? + 511$$

$$? = 350$$

$$12. \left[\frac{6}{7} \text{ of } 441 + \frac{9}{16} \text{ of } 384 \right] - \sqrt[3]{17576} = ? + 449$$

$$\sqrt[3]{17576} = 26$$

$$\left[\frac{6}{7} \times 441 + \frac{9}{16} \times 384 \right] - 26 = ? + 449$$

$$[378 + 216] - 26 = ? + 449$$

$$594 - 26 = ? + 449$$

$$568 = ? + 449$$

$$? = 119$$

$$13. 83.33\% \text{ of } 612 + \sqrt{2704} + 2^3 \times 11 = ? \times 13$$

$$83.33\% = \frac{5}{6}, \sqrt{2704} = 52 \text{ and } 2^3 \times 11 = 88$$

$$\frac{5}{6} \times 612 + 52 + 88 = ? \times 13$$

$$510 + 52 + 88 = ? \times 13$$

$$650 = ? \times 13$$

$$? = 50$$

$$14. \sqrt{?} + 75\% \text{ of } 360 = 16^2 + 100$$

$$75\% \text{ of } 360 = 270 \text{ and } 16^2 = 256$$

$$\sqrt{?} + 270 = 256 + 100$$

$$\sqrt{?} + 270 = 356$$

$$\sqrt{?} = 86$$

$$? = 7396$$

$$15. \left(\frac{3}{8} \text{ of } 640 \div 12 \right) + \frac{5}{9} \text{ of } 216 + 6^3 \div 9 = ?$$

$$6^3 = 216$$

$$\left(\frac{3}{8} \times 640 \div 12 \right) + \frac{5}{9} \times 216 + 216 \div 9 = ?$$

$$240 \div 12 + 120 + 24 = ?$$

$$20 + 120 + 24 = ?$$

$$? = 164$$

$$16. ? - 12.5\% \text{ of } 928 = \frac{4}{9} \text{ of } 810 + 13^2$$

$$12.5\% = \frac{1}{8} \text{ and } 13^2 = 169$$

$$? - \frac{1}{8} \times 928 = \frac{4}{9} \times 810 + 169$$

$$? - 116 = 360 + 169$$

$$? - 116 = 529$$

$$? = 645$$

$$17. 26 \times 18 - 14 \times 12 + 56.25\% \text{ of } 320 = ? \times 10$$

$$56.25\% = \frac{9}{16}$$

$$26 \times 18 - 14 \times 12 + \frac{9}{16} \times 320 = ? \times 10$$

$$468 - 168 + 180 = ? \times 10$$

$$480 = ? \times 10$$

$$? = 48$$

$$18. \frac{7}{11} \text{ of } 704 + \frac{5}{12} \text{ of } 720 - 68.75\% \text{ of } 576 = ? + 74$$

$$68.75\% = \frac{11}{16}$$

$$\frac{7}{11} \times 704 + \frac{5}{12} \times 720 - \frac{11}{16} \times 576 = ? + 74$$

$$448 + 300 - 396 = ? + 74$$

$$352 = ? + 74$$

$$? = 278$$

$$19. (\sqrt{7056} + \sqrt[3]{15625}) \times 5 - 37.5\% \text{ of } 480 = ?$$

$$\sqrt{7056} = 84, \sqrt[3]{15625} = 25 \text{ and } 37.5\% = \frac{3}{8}$$

$$(84 + 25) \times 5 - \frac{3}{8} \times 480 = ?$$

$$109 \times 5 - 180 = ?$$

$$545 - 180 = ?$$

$$? = 365$$

$$20. (72.72\% \text{ of } 605 + 32.5\% \text{ of } 800) \div ? = \sqrt{2025} + 5^2$$

$$72.72\% = \frac{8}{11}, 32.5\% = \frac{13}{40} \text{ and } \sqrt{2025} = 45$$

$$(\frac{8}{11} \times 605 + \frac{13}{40} \times 800) \div ? = 45 + 25$$

$$(440 + 260) \div ? = 70$$

$$700 \div ? = 70$$

$$? = 10$$

2. ARITHMETIC QUESTIONS

Q1: A motorboat takes 6 hours to cover a certain distance downstream and 10 hours to cover the same distance upstream. If the speed of the stream is 2 km/hr, find $4D$, where D is the distance covered one way.

एक मोटरबोट धारा के अनुकूल एक निश्चित दूरी 6 घंटे में और वही दूरी धारा के प्रतिकूल 10 घंटे में तय करती है। यदि धारा की गति 2 किमी/घंटा है, तो $4D$ ज्ञात कीजिए, जहां D एक तरफ की दूरी है।

- (A) 200
- (B) 220
- (C) 260
- (D) 300
- (E) 240

Q2: A set of five consecutive multiples of 8 has an average of 72. Another series consists of four consecutive even numbers, where the lowest even number is equal to the third multiple of the first series. Find the highest even number of the second series.

8 के पांच क्रमागत गुणजों के एक समूह का औसत 72 है। एक दूसरी श्रृंखला में चार क्रमागत सम संख्याएं हैं, जिनमें सबसे छोटी सम संख्या पहली श्रृंखला के तीसरे गुणज के बराबर है। दूसरी श्रृंखला की सबसे बड़ी सम संख्या ज्ञात कीजिए।

- (A) 74
- (B) 78
- (C) 82
- (D) 80
- (E) 76

Q3: The sum of the present ages of Person L and Person M is 84 years, and the sum of the present ages of Person M and Person N is 68 years. If the age of Person N after 8 years is 40 years, and the ratio of Person L's age after 6 years to Person M's age 4 years ago is $27 : X$, find X .

व्यक्ति L और व्यक्ति M की वर्तमान आयु का योग 84 वर्ष है, और व्यक्ति M और व्यक्ति N की वर्तमान आयु का योग 68 वर्ष है। यदि 8 वर्ष बाद व्यक्ति N की आयु 40 वर्ष होगी, और 6 वर्ष बाद व्यक्ति L की आयु तथा 4 वर्ष पहले व्यक्ति M की आयु का अनुपात $27 : X$ है, तो X ज्ञात कीजिए।

- (A) 12
- (B) 18
- (C) 20
- (D) 16
- (E) 14

Q4: A train is moving at a speed of 108 km/hr. It crosses a cyclist moving at 36 km/hr in the same direction in 15 seconds. The same train crosses a platform in 25 seconds. Find the length of the platform.

एक ट्रेन 108 किमी/घंटा की गति से चल रही है। यह समान दिशा में 36 किमी/घंटा की गति से चल रहे एक साइकिल सवार को 15 सेकंड में पार करती है। वही ट्रेन एक प्लेटफॉर्म को 25 सेकंड में पार करती है। प्लेटफॉर्म की लंबाई ज्ञात कीजिए।

- (A) 450 meters
- (B) 420 meters
- (C) 480 meters
- (D) 500 meters
- (E) 540 meters

Q5: A person invested Rs. X in Scheme A at 15% per annum for 2 years on simple interest. The amount obtained from Scheme A was reinvested in Scheme B at 10% per annum for 2 years on compound interest. If the difference between the interest earned from Scheme A and Scheme B is Rs. 162, find X .

एक व्यक्ति ने योजना A में Rs. X को 2 वर्षों के लिए 15% वार्षिक साधारण ब्याज पर निवेश किया। योजना A से प्राप्त राशि को योजना B में 2 वर्षों के लिए 10% वार्षिक चक्रवृद्धि ब्याज पर फिर से निवेश किया गया। यदि योजना A और योजना B से प्राप्त ब्याज के बीच अंतर Rs. 162 है, तो X ज्ञात कीजिए।

- (A) Rs. 5400
- (B) Rs. 4800
- (C) Rs. 6000

- (D) Rs. 7200
(E) Rs. 8000

Q6: A container has milk and water in the ratio of 7:y. When 16 liters of milk and 30 liters of water are added to the container, the ratio of milk to water becomes 10:9. Find the value of y. The initial quantity of the mixture is 144 liters.

एक बर्तन में दूध और पानी का अनुपात 7:y है। जब बर्तन में 16 लीटर दूध और 30 लीटर पानी मिलाया जाता है, तो दूध और पानी का अनुपात 10:9 हो जाता है। y का मान ज्ञात कीजिए। मिश्रण की प्रारंभिक मात्रा 144 लीटर है।

- (A) 4
(B) 7
(C) 6
(D) 5
(E) 3

Q7: The selling price of an article is Rs. 900, and the shopkeeper incurs a loss of 10%. If the same article is marked 50% above its cost price and the shopkeeper earns a profit of Rs. 200, find the discount percentage allowed by the shopkeeper.

एक वस्तु का विक्रय मूल्य Rs. 900 है और दुकानदार को 10% की हानि होती है। यदि उसी वस्तु को उसके क्रय मूल्य से 50% अधिक अंकित किया जाता है और दुकानदार Rs. 200 का लाभ कमाता है, तो दुकानदार द्वारा दी गई छूट प्रतिशत ज्ञात कीजिए।

- (A) 15%
(B) 20%
(C) 18%
(D) 25%
(E) 10%

Q8: A rectangular park has its length four times its breadth. A uniform pathway of width 2 meters runs inside the park along its boundary. If the area of the pathway is 264 sq. meters, find the length of the park.

एक आयताकार पार्क की लंबाई उसकी चौड़ाई की चार गुना है। पार्क की सीमा के अंदर 2 मीटर चौड़ा एक समान पथ बना है। यदि पथ का क्षेत्रफल 264 वर्ग मीटर है, तो पार्क की लंबाई ज्ञात कीजिए।

- (A) 48 meters
(B) 64 meters
(C) 60 meters
(D) 52 meters

(E) 56 meters

Q9: 4 men and 8 women can complete a work in 18 days. 7 men and 9 women can complete the same work in 12 days. If X men and 18 women can complete the work in 10 days, find X.

4 पुरुष और 8 महिलाएँ किसी कार्य को 18 दिनों में पूरा कर सकते हैं। 7 पुरुष और 9 महिलाएँ उसी कार्य को 12 दिनों में पूरा कर सकते हैं। यदि x पुरुष और 18 महिलाएँ उस कार्य को 10 दिनों में पूरा कर सकते हैं, तो x ज्ञात कीजिए।

- (A) 4
- (B) 9
- (C) 6
- (D) 8
- (E) 5

Q10: Rakesh invested Rs. 12000 and Sandeep invested Rs. 15000 in a business. After 4 months, Rakesh withdrew 25% of his initial investment. At the same time, Mahesh joined the business with an amount equal to 40% of Sandeep's initial investment. At the end of one year, the total profit was Rs. 87000. Find Rakesh's profit share.

राकेश ने Rs. 12000 और संदीप ने Rs. 15000 एक व्यवसाय में निवेश किए। 4 महीने बाद, राकेश ने अपने प्रारंभिक निवेश का 25% निकाल लिया। उसी समय, महेश संदीप के प्रारंभिक निवेश के 40% के बराबर राशि के साथ व्यवसाय में शामिल हुआ। एक वर्ष के अंत में कुल लाभ Rs. 87000 था। राकेश का लाभांश ज्ञात कीजिए।

- (A) Rs. 24000
- (B) Rs. 30000
- (C) Rs. 36000
- (D) Rs. 27000
- (E) Rs. 33000

Q11: Three men can complete a work in 20 days, while four women can complete the same work in 30 days. Find the time taken by four men and four women together to complete the whole work.

तीन पुरुष एक कार्य को 20 दिनों में पूरा कर सकते हैं, जबकि चार महिलाएँ उसी कार्य को 30 दिनों में पूरा कर सकती हैं। चार पुरुष और चार महिलाएँ मिलकर पूरे कार्य को कितने दिनों में पूरा करेंगी?

- (A) 8 days
- (B) 10 days

- (C) 12 days
- (D) 15 days
- (E) 9 days

Q12: The ratio of A to the sum of B and C is 3:5. The ratio of B to the sum of A and C is 2:3, and the ratio of C to the sum of A and B is 9:31. If $A + B + C = 800$, find the sum of the terms of the ratio of A to C.

A का B और C के योग से अनुपात 3:5 है। B का A और C के योग से अनुपात 2:3 है, और C का A और B के योग से अनुपात 9:31 है। यदि $A + B + C = 800$ है, तो A और C के अनुपात के पदों का योग ज्ञात कीजिए।

- (A) 7
- (B) 8
- (C) 9
- (D) 10
- (E) 11

Q13: A student scores 60% marks in an exam and passes by 50 marks. If he scores 35% marks in the same exam, he fails by x marks. If the maximum marks of the exam are 400, find 120% of x .

एक विद्यार्थी किसी परीक्षा में 60% अंक प्राप्त करता है और 50 अंकों से उत्तीर्ण होता है। यदि वह उसी परीक्षा में 35% अंक प्राप्त करता है, तो वह x अंकों से अनुत्तीर्ण हो जाता है। यदि परीक्षा के अधिकतम अंक 400 हैं, तो x का 120% ज्ञात कीजिए।

- (A) 50
- (B) 55
- (C) 60
- (D) 65
- (E) 70

Q14: A vessel contains 80 ml of a mixture of two liquids X and Y in the ratio of 3:5. If 40 ml of liquid Z is added to the vessel and then 24 ml of the resultant mixture is removed, find the sum of the terms of the ratio of liquid X to liquid Z in the final mixture.

एक बर्तन में दो द्रव X और Y का 80 ml मिश्रण 3:5 के अनुपात में है। यदि बर्तन में 40 ml द्रव Z मिलाया जाता है और फिर परिणामी मिश्रण में से 24 ml मिश्रण निकाल लिया जाता है, तो अंतिम मिश्रण में द्रव X और द्रव Z के अनुपात के पदों का योग ज्ञात कीजिए।

- (A) 5
- (B) 6

- (C) 7
(D) 8
(E) 9

Q15: Among nine consecutive even numbers, the average of the fourth largest number and the third smallest number is 95. Find the largest number.

नौ क्रमागत सम संख्याओं में चौथी सबसे बड़ी संख्या और तीसरी सबसे छोटी संख्या का औसत 95 है। सबसे बड़ी संख्या ज्ञात कीजिए।

- (A) 100
(B) 102
(C) 104
(D) 106
(E) 108

Q16: The difference between the upstream speed and downstream speed of a boat is 16 km/hr. The ratio of the speed of the boat in still water to the speed of the stream is 5 : 1. If the boat takes a total of 5 hours to cover D km upstream and D km downstream, find D.

एक नाव की धारा के विपरीत चाल और धारा के अनुकूल चाल के बीच का अंतर 16 किमी/घंटा है। स्थिर जल में नाव की चाल और धारा की चाल का अनुपात 5 : 1 है। यदि नाव D किमी धारा के विपरीत और D किमी धारा के अनुकूल दूरी तय करने में कुल 5 घंटे लेती है, तो D ज्ञात कीजिए।

- (A) 72
(B) 84
(C) 96
(D) 108
(E) 120

Q17: Yatin, Bhavesh and Lokesh invested Rs. 300, Rs. 450 and Rs. X respectively in a business. After 5 months, Lokesh withdrew Rs. 100, and after 3 more months, Yatin added Rs. 150 to his initial investment. At the end of the year, the combined profit share of Yatin and Bhavesh was Rs. 4800 out of the total profit of Rs. 6550. Find the value of X.

यतिन, भावेश और लोकेश ने एक व्यवसाय में क्रमशः Rs. 300, Rs. 450 और Rs. X का निवेश किया। 5 महीने बाद लोकेश ने Rs. 100 निकाल लिए और उसके 3 महीने बाद यतिन ने अपने शुरुआती निवेश में Rs. 150 और जोड़ दिए। वर्ष के अंत में कुल Rs. 6550 के लाभ में से यतिन और भावेश का संयुक्त लाभांश Rs. 4800 था। X का मान ज्ञात कीजिए।

- (A) 250
- (B) 300
- (C) 400
- (D) 350
- (E) 450

Q18: A person invested Rs. X in Scheme A at compound interest. The amount became Rs. 5290 after 2 years at 15% per annum. He invested Rs. Y in Scheme B at simple interest at 20% per annum for 2 years and received Rs. 200 more interest than the interest received from Scheme A after the first year. Find Y.

एक व्यक्ति ने योजना A में चक्रवृद्धि ब्याज पर Rs. X निवेश किए। 15% वार्षिक दर से 2 वर्षों बाद राशि Rs. 5290 हो गई। उसने योजना B में साधारण ब्याज पर 20% वार्षिक दर से 2 वर्षों के लिए Rs. Y निवेश किए और उसे योजना A से पहले वर्ष में प्राप्त ब्याज से Rs. 200 अधिक ब्याज मिला। Y ज्ञात कीजिए।

- (A) Rs. 1800
- (B) Rs. 2000
- (C) Rs. 2200
- (D) Rs. 2400
- (E) Rs. 2600

Q19: The ratio of the present age of a son to his father is 3 : 7. Six years hence, the father's age will be twice the son's age at that time. Find the father's age six years ago.

पुत्र और उसके पिता की वर्तमान आयु का अनुपात 3 : 7 है। 6 वर्ष बाद पिता की आयु उस समय पुत्र की आयु की दोगुनी होगी। 6 वर्ष पहले पिता की आयु ज्ञात कीजिए।

- (A) 30 years
- (B) 34 years
- (C) 36 years
- (D) 38 years
- (E) 40 years

Q20: A shopkeeper marked an article 20% above its cost price and allowed a discount of Rs. 150. If he sold the article for Rs. P and suffered a loss of 5%, find the value of P.

एक दुकानदार ने एक वस्तु का अंकित मूल्य उसके क्रय मूल्य से 20% अधिक रखा और Rs. 150 की छूट दी। यदि उसने वस्तु को Rs. P में बेचा और उसे 5% की हानि हुई, तो P का मान ज्ञात कीजिए।

- (A) Rs. 520
- (B) Rs. 540
- (C) Rs. 560
- (D) Rs. 580
- (E) Rs. 570

Answers:

1. E	11.B
2. B	12.B
3. D	13.C
4. A	14.C
5. C	15.C
6. D	16.C
7. B	17.D
8. E	18.B
9. C	19.C
10.B	20.E

Solutions:

1. Option E

Let the distance covered one way be D km.

Downstream time = 6 hours.

Upstream time = 10 hours.

Downstream speed = $D/6$ km/hr.

Upstream speed = $D/10$ km/hr.

In boat and stream questions:

Downstream speed - Upstream speed = $2 \times$ Speed of stream.

So,

$$D/6 - D/10 = 2 \times 2$$

$$D/6 - D/10 = 4$$

LCM of 6 and 10 = 30.

$$5D/30 - 3D/30 = 4$$

$$2D/30 = 4$$

$$D/15 = 4$$

$$D = 4 \times 15$$

$$D = 60 \text{ km.}$$

Required value = $4D$.

$$4D = 4 \times 60 = 240.$$

Therefore, the value of $4D$ is 240.

2. Option B

There are five consecutive multiples of 8.

In five consecutive numbers, the average is the middle number.

So, the third multiple of the first series = 72.

The lowest even number of the second series is equal to 72.

Four consecutive even numbers are:

72, 74, 76, 78.

The highest even number = 78.

Therefore, the highest even number of the second series is 78.

3. Option D

Age of Person N after 8 years = 40 years.

So, present age of Person N = $40 - 8 = 32$ years.

Present ages of Person M and Person N together = 68 years.

So, present age of Person M = $68 - 32 = 36$ years.

Present ages of Person L and Person M together = 84 years.

So, present age of Person L = $84 - 36 = 48$ years.

Person L's age after 6 years = $48 + 6 = 54$ years.

Person M's age 4 years ago = $36 - 4 = 32$ years.

Given ratio = $54 : 32$.

Simplify $54 : 32$ by dividing both terms by 2.

$$54 : 32 = 27 : 16.$$

$$\text{So, } 27 : X = 27 : 16.$$

Therefore, $X = 16$.

4. Option A

Speed of train = 108 km/hr.

Speed of train in m/sec = $108 \times 5/18 = 30$ m/sec.

Speed of cyclist = 36 km/hr.

Speed of cyclist in m/sec = $36 \times \frac{5}{18} = 10$ m/sec.

Both are moving in the same direction.

Relative speed = $30 - 10 = 20$ m/sec.

The train crosses the cyclist in 15 seconds.

So, length of train = Relative speed $\times$ Time.

Length of train = $20 \times 15 = 300$ meters.

The same train crosses a platform in 25 seconds.

Distance covered in 25 seconds = $30 \times 25 = 750$ meters.

While crossing a platform:

Length of train + Length of platform = 750.

So, length of platform = $750 - 300 = 450$ meters.

Therefore, the length of the platform is 450 meters.

5. Option C

Principal in Scheme A = Rs. X.

Rate = 15% per annum.

Time = 2 years.

Simple interest from Scheme A = $X \times 15 \times \frac{2}{100}$.

Simple interest from Scheme A = $\frac{30X}{100} = \frac{3X}{10}$.

Amount obtained from Scheme A = $X + \frac{3X}{10}$.

Amount obtained from Scheme A = $\frac{13X}{10}$.

This amount is reinvested in Scheme B at 10% compound interest for 2 years.

Compound interest factor for 2 years at 10% = $\frac{110}{100} \times \frac{110}{100}$.

Amount after Scheme B = $\frac{13X}{10} \times \frac{110}{100} \times \frac{110}{100}$.

Compound interest from Scheme B = $\frac{13X}{10} \times \frac{21}{100}$.

Compound interest from Scheme B = $\frac{273X}{1000}$.

Difference between interests = Rs. 162.

So,

$$\frac{3X}{10} - \frac{273X}{1000} = 162$$

$$\frac{300X}{1000} - \frac{273X}{1000} = 162$$

$$\frac{27X}{1000} = 162$$

$$X = 162 \times \frac{1000}{27}$$

$$X = 6000.$$

Therefore, the value of X is Rs. 6000.

6. Option D

Initial quantity of mixture = 144 liters

Initial ratio of milk to water = 7:y

Initial milk = $144 \times 7/(7 + y)$

Initial water = $144 \times y/(7 + y)$

After adding 16 liters milk and 30 liters water:

Milk = $144 \times 7/(7 + y) + 16$

Water = $144y/(7 + y) + 30$

New ratio = 10:9

So:

$[144 \times 7/(7 + y) + 16] : [144y/(7 + y) + 30] = 10 : 9$

$9[1008/(7 + y) + 16] = 10[144y/(7 + y) + 30]$

$9072/(7 + y) + 144 = 1440y/(7 + y) + 300$

$9072/(7 + y) - 1440y/(7 + y) = 300 - 144$

$(9072 - 1440y)/(7 + y) = 156$

$9072 - 1440y = 156(7 + y)$

$9072 - 1440y = 1092 + 156y$

$9072 - 1092 = 1440y + 156y$

$7980 = 1596y$

$y = 7980/1596$

$y = 5$

7. Option B

Selling price of the article = Rs. 900

Loss = 10%

So, selling price = 90% of cost price

Cost price = $900 \times 100/90$

Cost price = Rs. 1000

Now, the article is marked 50% above its cost price.

Marked price = 150% of 1000

Marked price = $150 \times 1000/100$

Marked price = Rs. 1500

The shopkeeper earns a profit of Rs. 200.

New selling price = Cost price + Profit

New selling price = $1000 + 200$

New selling price = Rs. 1200

Discount = Marked price - New selling price

Discount = $1500 - 1200$

Discount = Rs. 300

Discount percentage = $300 \times 100/1500$

Discount percentage = 20%

8. Option E

Let the breadth of the park be x meters.

Length of the park = $4x$ meters

Width of the pathway = 2 meters

Inner length = $4x - 2 \times 2 = 4x - 4$

Inner breadth = $x - 2 \times 2 = x - 4$

Area of pathway = Outer area - Inner area

$$264 = 4x \times x - (4x - 4)(x - 4)$$

$$264 = 4x^2 - (4x^2 - 16x - 4x + 16)$$

$$264 = 4x^2 - (4x^2 - 20x + 16)$$

$$264 = 20x - 16$$

$$20x = 264 + 16$$

$$20x = 280$$

$$x = 280/20$$

$$x = 14$$

Breadth = 14 meters

Length = 4×14

Length = 56 meters

Therefore, the length of the park is 56 meters.

9. Option C

Let the 1-day work of one man be m .

Let the 1-day work of one woman be w .

4 men and 8 women complete the work in 18 days.

Total work = $18(4m + 8w)$

7 men and 9 women complete the same work in 12 days.

Total work = $12(7m + 9w)$

So:

$$18(4m + 8w) = 12(7m + 9w)$$

$$72m + 144w = 84m + 108w$$

$$144w - 108w = 84m - 72m$$

$$36w = 12m$$

$$m = 3w$$

Total work = $18(4m + 8w)$

$$\text{Total work} = 18(4 \times 3w + 8w)$$

$$\text{Total work} = 18(12w + 8w)$$

$$\text{Total work} = 18 \times 20w$$

$$\text{Total work} = 360w$$

Now, X men and 18 women complete the work in 10 days.

$$10(Xm + 18w) = 360w$$

$$10(3Xw + 18w) = 360w$$

$$3Xw + 18w = 36w$$

$$3X + 18 = 36$$

$$3X = 18$$

$$X = 18/3$$

$$X = 6$$

10. Option B

Profit is divided in the ratio of capital $\times$ time.

Rakesh invested Rs. 12000 for the first 4 months.

After 4 months, Rakesh withdrew 25% of Rs. 12000.

$$\text{Amount withdrawn} = 25 \times 12000/100$$

$$\text{Amount withdrawn} = \text{Rs. } 3000$$

$$\text{Rakesh's new investment} = 12000 - 3000$$

$$\text{Rakesh's new investment} = \text{Rs. } 9000$$

$$\text{Rakesh's capital-months} = 12000 \times 4 + 9000 \times 8$$

$$\text{Rakesh's capital-months} = 48000 + 72000$$

$$\text{Rakesh's capital-months} = 120000$$

Sandeep invested Rs. 15000 for 12 months.

$$\text{Sandeep's capital-months} = 15000 \times 12$$

$$\text{Sandeep's capital-months} = 180000$$

Mahesh joined after 4 months with 40% of Sandeep's initial investment.

$$\text{Mahesh's investment} = 40 \times 15000/100$$

$$\text{Mahesh's investment} = \text{Rs. } 6000$$

Mahesh invested for remaining 8 months.

$$\text{Mahesh's capital-months} = 6000 \times 8$$

$$\text{Mahesh's capital-months} = 48000$$

$$\text{Profit ratio} = 120000 : 180000 : 48000$$

$$\text{Profit ratio} = 10 : 15 : 4$$

$$\text{Total ratio parts} = 10 + 15 + 4 = 29$$

$$\text{Total profit} = \text{Rs. } 87000$$

$$\text{Rakesh's share} = 87000 \times 10/29$$

Rakesh's share = Rs. 30000

11. Option B

Take the total work as 60 units, because 3 men complete the work in 20 days.

Work done by 3 men in 1 day = $60/20 = 3$ units

So, work done by 1 man in 1 day = $3/3 = 1$ unit

Work done by 4 women in 1 day = $60/30 = 2$ units

So, work done by 1 woman in 1 day = $2/4 = 1/2$ unit

Work done by 4 men in 1 day = $4 \times 1 = 4$ units

Work done by 4 women in 1 day = $4 \times 1/2 = 2$ units

Total work done by 4 men and 4 women in 1 day = $4 + 2 = 6$ units

Time taken = Total work/Total work done in 1 day

Time taken = $60/6 = 10$ days

12. Option B

Let $A + B + C = 800$.

Given, $A : (B + C) = 3:5$.

So, A is 3 parts out of total $3 + 5 = 8$ parts.

$A = 800 \times 3/8 = 300$

Given, $C : (A + B) = 9:31$.

So, C is 9 parts out of total $9 + 31 = 40$ parts.

$C = 800 \times 9/40 = 180$

Now, ratio of A to C = $300:180$.

Simplify the ratio:

$300:180 = 5:3$

Sum of terms of the ratio = $5 + 3 = 8$

13. Option C

Maximum marks = 400.

Marks scored at 60% = $400 \times 60/100 = 240$

The student passes by 50 marks.

So, passing marks = $240 - 50 = 190$

Marks scored at 35% = $400 \times 35/100 = 140$

He fails by x marks.

So, $x = \text{Passing marks} - \text{marks scored}$

$x = 190 - 140 = 50$

Required value = 120% of x

Required value = $50 \times 120/100$

Required value = 60

14. Option C

Total initial mixture = 80 ml.

Ratio of liquid X to liquid Y = 3:5.

Total parts = $3 + 5 = 8$

Quantity of X = $80 \times 3/8 = 30$ ml

Quantity of Y = $80 \times 5/8 = 50$ ml

Now 40 ml of liquid Z is added.

Quantity of Z = 40 ml

Total mixture = $80 + 40 = 120$ ml

Now 24 ml of the resultant mixture is removed.

Removed quantity = $24/120 = 1/5$ of the mixture

So, remaining quantity of each liquid = $4/5$ of its quantity.

Final quantity of X = $30 \times 4/5 = 24$ ml

Final quantity of Z = $40 \times 4/5 = 32$ ml

Ratio of X to Z = 24:32

Simplify the ratio:

24:32 = 3:4

Sum of terms = $3 + 4 = 7$

15. Option C

Let the nine consecutive even numbers be:

$a, a + 2, a + 4, a + 6, a + 8, a + 10, a + 12, a + 14, a + 16$

Third smallest number = $a + 4$

Fourth largest number = $a + 10$

Average of these two numbers = 95

So, $[(a + 4) + (a + 10)]/2 = 95$

$(2a + 14)/2 = 95$

$a + 7 = 95$

$a = 95 - 7 = 88$

Largest number = $a + 16$

Largest number = $88 + 16 = 104$

16. Option C

Let the speed of the boat in still water be $5x$ and the speed of the stream be x .

Downstream speed = $5x + x = 6x$

Upstream speed = $5x - x = 4x$

Difference between downstream and upstream speeds = 16 km/hr

So, $6x - 4x = 16$

$2x = 16$

$x = 16/2 = 8$

Speed of boat in still water = $5 \times 8 = 40$ km/hr

Speed of stream = 8 km/hr

Downstream speed = $40 + 8 = 48$ km/hr

Upstream speed = $40 - 8 = 32$ km/hr

The boat covers D km upstream and D km downstream in 5 hours.

So, $D/32 + D/48 = 5$

LCM of 32 and 48 = 96

$3D/96 + 2D/96 = 5$

$5D/96 = 5$

$D = 5 \times 96/5$

$D = 96$

17. Option D

In partnership, profit is divided in the ratio of capital $\times$ time.

Yatin invested Rs. 300 for the first 8 months.

After 8 months, he added Rs. 150.

So, his investment for the last 4 months = $300 + 150 =$ Rs. 450

Yatin's capital-time value = $300 \times 8 + 450 \times 4$

Yatin's capital-time value = $2400 + 1800 = 4200$

Bhavesh invested Rs. 450 for 12 months.

Bhavesh's capital-time value = $450 \times 12 = 5400$

Combined capital-time value of Yatin and Bhavesh = $4200 + 5400 = 9600$

Combined profit share of Yatin and Bhavesh = Rs. 4800

Total profit = Rs. 6550

So, their combined share ratio = $4800/6550 = 96/131$

This means 96 parts = 9600

1 part = $9600/96 = 100$

Total capital-time value = $131 \times 100 = 13100$

Lokesh's capital-time value = $13100 - 9600 = 3500$

Lokesh invested Rs. X for the first 5 months.

After 5 months, he withdrew Rs. 100.

So, his investment for the next 7 months = $X - 100$

Lokesh's capital-time value = $5X + 7(X - 100)$

$$3500 = 5X + 7X - 700$$

$$3500 = 12X - 700$$

$$12X = 3500 + 700$$

$$12X = 4200$$

$$X = 4200/12 = 350$$

18. Option B

Amount in Scheme A after 2 years = Rs. 5290

Rate of compound interest = 15% per annum

Amount after 2 years = Principal $\times 115/100 \times 115/100$

$$\text{So, } X \times 115/100 \times 115/100 = 5290$$

$$X \times 13225/10000 = 5290$$

$$X = 5290 \times 10000/13225$$

$$X = \text{Rs. } 4000$$

Interest received from Scheme A after the first year = $4000 \times 15/100$

Interest = Rs. 600

Interest from Scheme B is Rs. 200 more than this.

So, interest from Scheme B = $600 + 200 = \text{Rs. } 800$

Scheme B gives simple interest at 20% per annum for 2 years.

Simple interest = $Y \times 20 \times 2/100$

$$800 = 40Y/100$$

$$800 = 2Y/5$$

$$Y = 800 \times 5/2$$

$$Y = \text{Rs. } 2000$$

19. Option C

Present age ratio of son and father = 3 : 7

Let son's present age = $3x$

Let father's present age = $7x$

After 6 years:

Son's age = $3x + 6$

Father's age = $7x + 6$

Given, father's age after 6 years will be twice the son's age.

$$\text{So, } 7x + 6 = 2(3x + 6)$$

$$7x + 6 = 6x + 12$$

$$7x - 6x = 12 - 6$$

$$x = 6$$

Father's present age = $7 \times 6 = 42$ years

Father's age 6 years ago = $42 - 6 = 36$ years

20. Option E

Let the cost price of the article be Rs. x .

Marked price is 20% above cost price.

Marked price = 120% of $x = 120x/100 = 6x/5$

Discount = Rs. 150

Selling price = Marked price - Discount

Selling price = $6x/5 - 150$

The shopkeeper suffered a loss of 5%.

So, selling price = 95% of cost price

Selling price = $95x/100 = 19x/20$

Now,

$$6x/5 - 150 = 19x/20$$

LCM of 5 and 20 = 20

$$24x/20 - 19x/20 = 150$$

$$5x/20 = 150$$

$$x/4 = 150$$

$$x = 600$$

Cost price = Rs. 600

Selling price $P = 95\%$ of 600

$$P = 600 \times 95/100$$

$$P = \text{Rs. } 570$$

3. Quadratic Equations

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

(a) $x > y$

(b) $x < y$

(c) $x = y$ or the relationship cannot be established

(d) $x \geq y$

(e) $x \leq y$

1.) I. $8x^2 - 50x + 77 = 0$
II. $12y^2 - 25y - 22 = 0$

2.) I. $3x^2 - 22x + 35 = 0$
II. $10y^2 - 7y - 12 = 0$

3.) I. $20x^2 - 83x + 77 = 0$
II. $5y^2 - 32y + 51 = 0$

4.) I. $6x^2 - 17x - 45 = 0$
II. $4y^2 + 5y - 21 = 0$

5.) I. $4x^2 - 19x + 21 = 0$
II. $3y^2 - 19y + 30 = 0$

6.) I. $4x^2 - 27x + 44 = 0$
II. $8y^2 - 14y - 15 = 0$

7.) I. $12x^2 - 7x - 10 = 0$
II. $3y^2 - 11y + 10 = 0$

8.) I. $8x^2 - 46x + 65 = 0$
II. $10y^2 - 19y - 15 = 0$

9.) I. $5x^2 + 2x - 39 = 0$
II. $3y^2 - 2y - 8 = 0$

10.) I. $15x^2 - 62x + 63 = 0$

II. $3y^2 - 16y + 21 = 0$

11.) I. $8x^2 - 54x + 91 = 0$

II. $18y^2 - 33y - 40 = 0$

12.) I. $20x^2 - 61x + 45 = 0$

II. $5y^2 - 21y + 22 = 0$

13.) I. $12x^2 - 79x + 130 = 0$

II. $4y^2 - 17y + 13 = 0$

14.) I. $8x^2 - 22x - 63 = 0$

II. $6y^2 - 5y - 21 = 0$

15.) I. $10x^2 - 73x + 133 = 0$

II. $3y^2 - 4y - 20 = 0$

16.) I. $12x^2 - 59x + 72 = 0$

II. $15y^2 - 91y + 136 = 0$

17.) I. $10x^2 - 17x - 11 = 0$

II. $15y^2 - 76y + 96 = 0$

18.) I. $10x^2 - 69x + 119 = 0$

II. $5y^2 - 22y + 17 = 0$

19.) I. $12x^2 - 31x - 26 = 0$

II. $5y^2 - 3y - 8 = 0$

20.) I. $12x^2 - 71x + 104 = 0$

II. $6y^2 - 7y - 20 = 0$

Answers:

1. D

2. A

3. B

4. C

5. E

6. A

7. B

8. D

9. C

10. E

11. A

12. B

13. D

14. C

15. A

16. E

17. B

18. D

19. C

20. A

Solutions:

$$(1) x = \frac{7}{2}, \frac{11}{4}$$
$$y = \frac{11}{4}, -\frac{2}{3}$$

$$(2) x = 5, \frac{7}{3}$$
$$y = \frac{3}{2}, -\frac{4}{5}$$

$$(3) x = \frac{11}{4}, \frac{7}{5}$$
$$y = \frac{17}{5}, 3$$

$$(4) x = \frac{9}{2}, -\frac{5}{3}$$
$$y = \frac{7}{4}, -3$$

$$(5) x = 3, \frac{7}{4}$$
$$y = \frac{10}{3}, 3$$

$$(6) x = 4, \frac{11}{4}$$
$$y = \frac{5}{2}, -\frac{3}{4}$$

$$(7) x = \frac{5}{4}, -\frac{2}{3}$$
$$y = 2, \frac{5}{3}$$

$$(8) x = \frac{13}{4}, \frac{5}{2}$$
$$y = \frac{5}{2}, -\frac{3}{5}$$

$$(9) x = \frac{13}{5}, -3$$
$$y = 2, -\frac{4}{3}$$

$$(10) x = \frac{7}{3}, \frac{9}{5}$$
$$y = 3, \frac{7}{3}$$

$$(11) x = \frac{7}{2}, \frac{13}{4}$$
$$y = \frac{8}{3}, -\frac{5}{6}$$

$$(12) x = \frac{9}{5}, \frac{5}{4}$$
$$y = \frac{11}{5}, 2$$

$$(13) x = \frac{10}{3}, \frac{13}{4}$$
$$y = \frac{13}{4}, 1$$

$$(14) x = \frac{9}{2}, -\frac{7}{4}$$
$$y = \frac{7}{3}, -\frac{3}{2}$$

$$(15) x = \frac{19}{5}, \frac{7}{2}$$
$$y = \frac{10}{3}, -2$$

$$(16) x = \frac{8}{3}, \frac{9}{4}$$
$$y = \frac{17}{5}, \frac{8}{3}$$

$$(17) x = \frac{11}{5}, -\frac{1}{2}$$
$$y = \frac{8}{3}, \frac{12}{5}$$

$$(18) \ x = \frac{7}{2}, \frac{17}{5}$$

$$y = \frac{17}{5}, 1$$

$$(19) \ x = \frac{13}{4}, -\frac{2}{3}$$

$$y = \frac{8}{5}, -1$$

$$(20) \ x = \frac{13}{4}, \frac{8}{3}$$

$$y = \frac{5}{2}, -\frac{4}{3}$$

4. WRONG NUMBER SERIES

(1) 4860, 4877, 4909, 4941, 4994, 5065

(a) 4941

(b) 4909

(c) 4994

(d) 5065

(e) None of these

(2) 374, 761, 1539, 3099, 6230, 12475

(a) 1539

(b) 761

(c) 12475

(d) 6230

(e) None of these

(3) 27846, 13932, 4652, 1170, 240, 50

(a) 13932

(b) 1170

(c) 50

(d) 240

(e) None of these

(4) 4980, 2530, 1300, 690, 365, 202.5

(a) 690

(b) 1300

(c) 365

(d) 2530

(e) None of these

(5) 1350, 1374, 1434, 1554, 1770, 2100

(a) 1434

(b) 1374

(c) 2100

(d) 1554

(e) None of these

(6) 2520, 2500, 2474, 2428, 2372, 2300

(a) 2474

(b) 2500

(c) 2428

(d) 2300

(e) None of these

(7) 620.5, 628, 648, 665.5, 695.5, 733

(a) 665.5

(b) 648

(c) 620.5

(d) 733

(e) None of these

(8) 720, 738, 765, 810, 892, 999

(a) 738

(b) 810

(c) 892

(d) 999

(e) None of these

(9) 326, 365, 406, 456, 494, 541

(a) 365

(b) 494

(c) 541

(d) 456

(e) None of these

(10) 186840, 62300, 15600, 3125, 560, 120

(a) 15600

(b) 62300

(c) 560

(d) 120

(e) None of these

(11) 144, 283, 839, 3350, 16685, 100085

(a) 16685

(b) 3350

(c) 839

(d) 144

(e) None of these

(12) 1400, 1382, 1346, 1286, 1202, 1070

(a) 1382

(b) 1346

(c) 1400

(d) 1202

(e) None of these

(13) 200, 290, 560, 1370, 4055, 14195

(a) 4055

(b) 560

(c) 1370

(d) 14195

(e) None of these

(14) 300, 421, 516, 602, 666, 715

(a) 421

(b) 666

(c) 516

(d) 715

(e) None of these

(15) 75, 224, 670, 2006, 6022, 18014

(a) 670

(b) 224

(c) 2006

(d) 18014

(e) None of these

(16) 768, 384, 576, 288, 426, 216

(a) 576

(b) 384

(c) 216

(d) 768

(e) None of these

(17) 930, 936, 956, 1002, 1070, 1180

(a) 1070

(b) 1002

(c) 936

(d) 1180

(e) None of these

(18) 990, 972, 945, 909, 856, 810

(a) 856

(b) 945

(c) 972

(d) 810

(e) None of these

(19) 300, 581, 1139, 2249, 4480, 8897

(a) 581

(b) 1139

(c) 8897

(d) 4480

(e) None of these

(20) 540.25, 548.75, 565.75, 596.25, 625.25, 667.75

(a) 625.25

(b) 548.75

(c) 596.25

(d) 667.75

(e) None of these

Answers:

(1) b

(2) d

(3) c

(4) a

(5) e

(6) a

(7) b

(8) c

(9) d

(10) e	(6) $-(4^2 + 4)$, $-(5^2 + 5)$, $-(6^2 + 6)$, $-(7^2 + 7)$, $-(8^2 + 8)$
(11) b	
(12) d	(7) +7.5, +15, +22.5, +30, +37.5
(13) a	(8) +18, +27, +45, +72, +117
(14) c	(9) $18^2 + 2$, $19^2 + 4$, $20^2 + 6$, $21^2 + 8$, $22^2 + 10$, $23^2 + 12$
(15) e	(10) $\div 3 + 20$, $\div 4 + 25$, $\div 5 + 30$, $\div 6 + 35$, $\div 7 + 40$
(16) e	
(17) b	(11) $\times 2 - 5$, $\times 3 - 10$, $\times 4 - 15$, $\times 5 - 20$, $\times 6 - 25$
(18) a	
(19) d	(12) $-(2 \times 9)$, $-(3 \times 12)$, $-(4 \times 15)$, $-(5 \times 18)$, $-(6 \times 21)$
(20) c	(13) $\times 1.5 - 10$, $\times 2 - 20$, $\times 2.5 - 30$, $\times 3 - 40$, $\times 3.5 - 50$
Solutions:	(14) $+11^2$, $+10^2$, $+9^2$, $+8^2$, $+7^2$
(1) +17, +26, +38, +53, +71 +9, +12, +15, +18	(15) $\times 3 - 1$, $\times 3 - 2$, $\times 3 - 4$, $\times 3 - 8$, $\times 3 - 16$
(2) $\times 2 + 13$, $\times 2 + 17$, $\times 2 + 21$, $\times 2 + 25$, $\times 2 + 29$	(16) $\div 2$, $\times 1.5$, $\div 2$, $\times 1.5$, $\div 2$
(3) $\div 2 + 9$, $\div 3 + 8$, $\div 4 + 7$, $\div 5 + 6$, $\div 6 + 5$	(17) $+(2 \times 3)$, $+(4 \times 5)$, $+(6 \times 7)$, $+(8 \times 9)$, $+(10 \times 11)$
(4) $\div 2 + 40$, $\div 2 + 35$, $\div 2 + 30$, $\div 2 + 25$, $\div 2 + 20$	(18) -18, -27, -36, -45, -54
(5) $+(3^3 - 3)$, $+(4^3 - 4)$, $+(5^3 - 5)$, $+(6^3 - 6)$, $+(7^3 - 7)$	(19) $\times 2 - 19$, $\times 2 - 23$, $\times 2 - 29$, $\times 2 - 31$, $\times 2 - 37$
	(20) +8.5, +17, +25.5, +34, +42.5

5. MISSING NUMBER SERIES

(1) 3200, 3185, 3155, 3110, ?, 2975

- (a) 3040
- (b) 3050
- (c) 3060
- (d) 3070
- (e) None of these

(2) 4, 9, 19, ?, 54, 79

- (a) 34
- (b) 32
- (c) 36
- (d) 38
- (e) None of these

(3) 35, 72, ?, 298, 601, 1208

- (a) 143
- (b) 145
- (c) 147
- (d) 149
- (e) None of these

(4) 4096, 1024, ?, 64, 16, 4

- (a) 252
- (b) 254
- (c) 256
- (d) 258
- (e) None of these

(5) 118, 157, 275, ?, 707, 1139

- (a) 430
- (b) 432
- (c) 434

(d) 436

(e) None of these

(6) 26, 33, 44, 57, ?, 93

- (a) 68
- (b) 70
- (c) 72
- (d) 74

(e) None of these

(7) 25, 80, 245, 740, 2225, ?

- (a) 6670
- (b) 6680
- (c) 6690
- (d) 6700
- (e) None of these

(8) 12.5, 15, 20, ?, 37.5, 50

- (a) 27.5
- (b) 28.5
- (c) 29.5
- (d) 30.5
- (e) None of these

(9) 875, 865, ?, 815, 775, 725

- (a) 835
- (b) 840
- (c) 845
- (d) 850
- (e) None of these

(10) 12, 38, 118, ?, 1088, 3274

- (a) 354
- (b) 356

- (c) 358
(d) 360
(e) None of these

(11) 210, 224, 245, 273, ?, 350

- (a) 306
(b) 308
(c) 310
(d) 312
(e) None of these

(12) 3888, 1296, 432, ?, 48, 16

- (a) 138
(b) 140
(c) 142
(d) 144
(e) None of these

(13) 640, 634, ?, 611, 590, 560

- (a) 625
(b) 627
(c) 629
(d) 631
(e) None of these

(14) 121, 130, 148, ?, 211, 256

- (a) 173
(b) 175
(c) 177
(d) 179
(e) None of these

(15) 42, 85, 171, ?, 687, 1375

- (a) 337
(b) 339
(c) 341
(d) 343
(e) None of these

(16) 5040, 840, 168, ?, 14, 7

- (a) 38
(b) 40
(c) 42
(d) 44
(e) None of these

(17) 77, 99, 143, ?, 407, 759

- (a) 227
(b) 229
(c) 231
(d) 233
(e) None of these

(18) 9, 26, 77, 230, ?, 2066

- (a) 685
(b) 687
(c) 689
(d) 691
(e) None of these

(19) 1450, 1470, 1510, 1570, ?, 1750

- (a) 1640
(b) 1650
(c) 1660
(d) 1670
(e) None of these

(20) 68, 80, 104, ?, 248, 440

- (a) 148
(b) 150
(c) 152
(d) 154
(e) None of these

Answers:

- (1) b
(2) a
(3) c

- (4) c
- (5) b
- (6) d
- (7) b
- (8) a
- (9) c
- (10) d
- (11) b
- (12) d
- (13) a
- (14) b
- (15) d
- (16) c
- (17) c
- (18) c
- (19) b
- (20) c

Solutions:

- (1) -15, -30, -45, -60, -75
- (2) +5, +10, +15, +20, +25
- (3) $\times 2 + 2$, $\times 2 + 3$, $\times 2 + 4$, $\times 2 + 5$, $\times 2 + 6$
- (4) $\div 4$, $\div 4$, $\div 4$, $\div 4$, $\div 4$
- (5) Sum of the previous two numbers

- (6) +7, +11, +13, +17, +19
- (7) $\times 3 + 5$, $\times 3 + 5$, $\times 3 + 5$, $\times 3 + 5$, $\times 3 + 5$
- (8) +2.5, +5, +7.5, +10, +12.5
- (9) -10, -20, -30, -40, -50
- (10) $\times 3 + 2$, $\times 3 + 4$, $\times 3 + 6$, $\times 3 + 8$, $\times 3 + 10$
- (11) +14, +21, +28, +35, +42
- (12) $\div 3$, $\div 3$, $\div 3$, $\div 3$, $\div 3$
- (13) $-(1^2 + 5)$, $-(2^2 + 5)$, $-(3^2 + 5)$, $-(4^2 + 5)$, $-(5^2 + 5)$
- (14) +9, +18, +27, +36, +45
- (15) $\times 2 + 1$, $\times 2 + 1$, $\times 2 + 1$, $\times 2 + 1$, $\times 2 + 1$
- (16) $\div 6$, $\div 5$, $\div 4$, $\div 3$, $\div 2$
- (17) +22, +44, +88, +176, +352
- (18) $\times 3 - 1$, $\times 3 - 1$, $\times 3 - 1$, $\times 3 - 1$, $\times 3 - 1$
- (19) +20, +40, +60, +80, +100
- (20) +12, +24, +48, +96, +192

6. DATA INTERPRETATION

Set 1: Read the following table carefully and answer the questions given below.
The table shows the total number of feedback forms received by five different food outlets and the percentage of feedback forms rated 4-star or above in each outlet.

निम्नलिखित तालिका को ध्यानपूर्वक पढ़िए और नीचे दिए गए प्रश्नों के उत्तर दीजिए।
तालिका पाँच अलग-अलग फूड आउटलेट्स द्वारा प्राप्त कुल फीडबैक फॉर्म और प्रत्येक आउटलेट में 4-स्टार या उससे अधिक रेटिंग वाले फीडबैक फॉर्म का प्रतिशत दर्शाती है।

Outlet / आउटलेट	Total feedback forms received / प्राप्त कुल फीडबैक फॉर्म	Percentage of feedback forms rated 4-star or above / 4-स्टार या उससे अधिक रेटिंग वाले फीडबैक फॉर्म का प्रतिशत
Jaipur Outlet / जयपुर आउटलेट	960	37.5%
Indore Outlet / इंदौर आउटलेट	1200	64%
Kochi Outlet / कोच्चि आउटलेट	1600	50%
Nashik Outlet / नासिक आउटलेट	1088	50%
Surat Outlet / सूरत आउटलेट	1280	25%

Q1: Find the average number of feedback forms rated below 4-star from Indore Outlet, Kochi Outlet and Nashik Outlet together.

इंदौर आउटलेट, कोच्चि आउटलेट और नासिक आउटलेट से 4-स्टार से कम रेटिंग वाले फीडबैक फॉर्मों की औसत संख्या ज्ञात कीजिए।

- (A) 568
- (B) 608
- (C) 584
- (D) 592
- (E) 616

Q2: If the male to female customer ratio in Nashik Outlet and Surat Outlet is 9:8 and 7:9 respectively, then the ratio of total male customers to total female customers in both outlets together is P:Q. Find P + Q.

यदि नासिक आउटलेट और सूरत आउटलेट में पुरुष और महिला ग्राहकों का अनुपात क्रमशः 9:8 और 7:9 है, तो दोनों आउटलेट्स में मिलाकर कुल पुरुष ग्राहकों और कुल

महिला ग्राहकों का अनुपात P:Q है। P + Q ज्ञात कीजिए।

- (A) 148
- (B) 152
- (C) 144
- (D) 156
- (E) 140

Q3: The number of feedback forms rated 4-star or above from Nashik Outlet is what percentage more or less than the number of feedback forms rated 4-star or above from Jaipur Outlet and Surat Outlet together?

नासिक आउटलेट से 4-स्टार या उससे अधिक रेटिंग वाले फीडबैक फॉर्म, जयपुर आउटलेट और सूरत आउटलेट से मिलाकर 4-स्टार या उससे अधिक रेटिंग वाले फीडबैक फॉर्मों से कितने प्रतिशत अधिक या कम हैं?

- (A) 25%
- (B) 15%
- (C) 10%
- (D) 30%
- (E) 20%

Q4: If the ratio of satisfied to dissatisfied customers in Kochi Outlet is 7:3 and all feedback forms rated 4-star or above were given by satisfied customers, then how many satisfied customers gave ratings below 4-star in Kochi Outlet?

यदि कोच्चि आउटलेट में संतुष्ट और असंतुष्ट ग्राहकों का अनुपात 7:3 है, तो कोच्चि आउटलेट में कितने संतुष्ट ग्राहकों ने 4-स्टार से कम रेटिंग दी?

- (A) 280
- (B) 360
- (C) 320
- (D) 400
- (E) 240

Q5: Find the difference between feedback forms rated below 4-star from Jaipur Outlet and Indore Outlet together and feedback forms rated 4-star or above from Kochi Outlet and Surat Outlet together.

जयपुर आउटलेट और इंदौर आउटलेट से मिलाकर 4-स्टार से कम रेटिंग वाले फीडबैक

फॉर्मों तथा कोच्चि आउटलेट और सूरत आउटलेट से मिलाकर 4-स्टार या उससे अधिक रेटिंग वाले फीडबैक फॉर्मों के बीच का अंतर ज्ञात कीजिए।

- (A) 96
(B) 88
(C) 104
(D) 72
(E) 80

Solutions:

Outlet / आउटलेट	4-star or above / 4-स्टार या अधिक	Below 4-star / 4-स्टार से कम
Jaipur Outlet / जयपुर आउटलेट	360	600
Indore Outlet / इंदौर आउटलेट	768	432
Kochi Outlet / कोच्चि आउटलेट	800	800
Nashik Outlet / नासिक आउटलेट	544	544
Surat Outlet / सूरत आउटलेट	320	960

Common values:

Jaipur Outlet: 4-star or above = 37.5% of 960 = $\frac{3}{8} \times 960 = 360$, Below 4-star = $960 - 360 = 600$

Indore Outlet: 4-star or above = 64% of 1200 = 768, Below 4-star = $1200 - 768 = 432$

Kochi Outlet: 4-star or above = 50% of 1600 = 800, Below 4-star = $1600 - 800 = 800$

Nashik Outlet: 4-star or above = 50% of 1088 = 544, Below 4-star = $1088 - 544 = 544$

Surat Outlet: 4-star or above = 25% of 1280 = 320, Below 4-star = $1280 - 320 = 960$

1. Option D

Below 4-star forms from Indore, Kochi and Nashik = $432 + 800 + 544 = 1776$

Average = $1776/3 = 592$

2. Option A

Male customers in Nashik = $1088 \times \frac{9}{17} = 576$

Female customers in Nashik = $1088 \times \frac{8}{17} = 512$

Male customers in Surat = $1280 \times \frac{7}{16} = 560$

Female customers in Surat = $1280 \times \frac{9}{16} = 720$

Required ratio = $(576 + 560):(512 + 720) = 1136:1232 = 71:77$

$P + Q = 71 + 77 = 148$

3. Option E

4-star or above forms from Nashik = 544

4-star or above forms from Jaipur and Surat = $360 + 320 = 680$

Difference = $680 - 544 = 136$

Required percentage = $136/680 \times 100 = 20\%$ less

4. Option C

Satisfied customers in Kochi = $7/10 \times 1600 = 1120$

4-star or above forms in Kochi = 800

Satisfied customers with below 4-star rating = $1120 - 800 = 320$

5. Option B

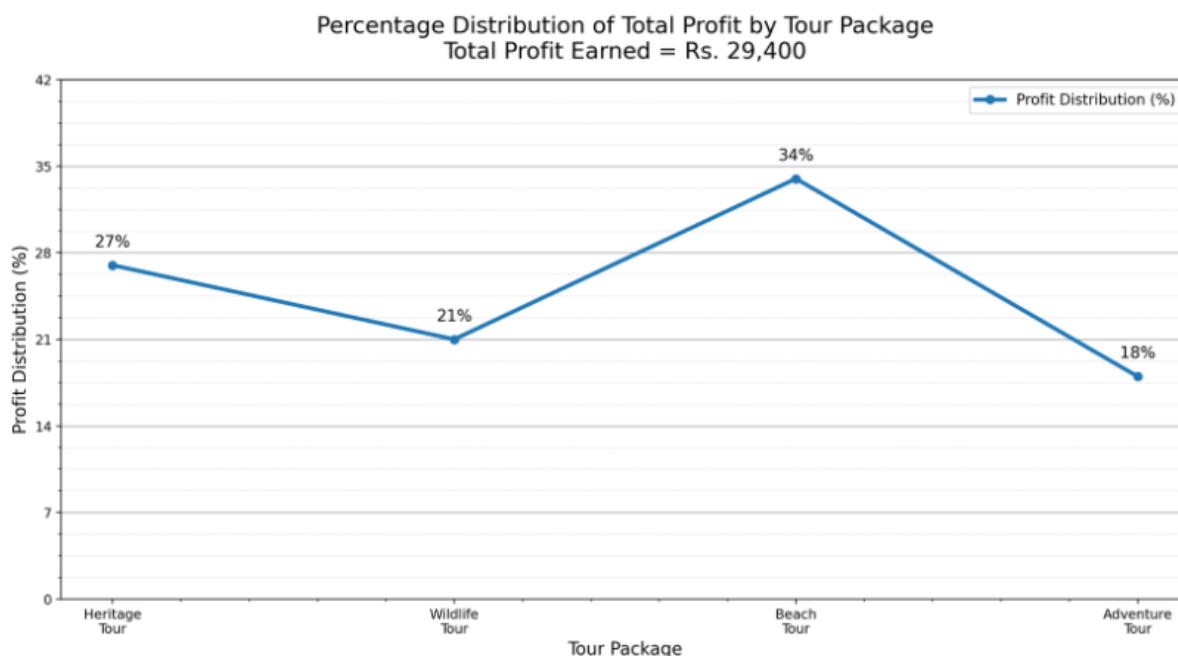
Below 4-star forms from Jaipur and Indore = $600 + 432 = 1032$

4-star or above forms from Kochi and Surat = $800 + 320 = 1120$

Required difference = $1120 - 1032 = 88$

Set 2: The line graph shows the percentage distribution of the total profit earned by a travel agency by selling four different tour packages: Heritage Tour, Wildlife Tour, Beach Tour and Adventure Tour. The total profit earned by the agency is Rs. 29,400. The table shows the total number of packages sold for each tour package. Study the data carefully and answer the following questions.

रेखा आलेख एक ट्रैवल एजेंसी द्वारा चार अलग-अलग टूर पैकेजों: हेरिटेज टूर, वाइल्डलाइफ टूर, बीच टूर और एडवेंचर टूर को बेचकर अर्जित कुल लाभ का प्रतिशत वितरण दर्शाता है। एजेंसी द्वारा अर्जित कुल लाभ Rs. 29,400 है। तालिका प्रत्येक टूर पैकेज के बेचे गए कुल पैकेजों की संख्या दर्शाती है। आंकड़ों का ध्यानपूर्वक अध्ययन करें और निम्नलिखित प्रश्नों के उत्तर दें।



Tour Package	Profit %	Packages Sold	Profit per Package
Heritage Tour	27%	126	Rs. 63
Wildlife Tour	21%	147	Rs. 42
Beach Tour	34%	119	Rs. 84
Adventure Tour	18%	189	Rs. 28

Q1: If the cost price of one Heritage Tour package is Rs. 735, then find the selling price of one Heritage Tour package.

यदि एक हेरिटेज टूर पैकेज का क्रय मूल्य Rs. 735 है, तो एक हेरिटेज टूर पैकेज का विक्रय मूल्य ज्ञात कीजिए।

- (A) Rs. 812
- (B) Rs. 785
- (C) Rs. 805
- (D) Rs. 798
- (E) Rs. 792

Q2: If the cost price of one Wildlife Tour package to one Beach Tour package is in the ratio 2 : 5, and the selling price of one Wildlife Tour package to one Beach Tour package is in the ratio 5 : 12, then find the selling price of one Wildlife Tour package.

यदि एक वाइल्डलाइफ टूर पैकेज और एक बीच टूर पैकेज के क्रय मूल्य का अनुपात 2 : 5 है, और एक वाइल्डलाइफ टूर पैकेज और एक बीच टूर पैकेज के विक्रय मूल्य का अनुपात 5 : 12 है, तो एक वाइल्डलाइफ टूर पैकेज का विक्रय मूल्य ज्ञात कीजिए।

- (A) Rs. 205
- (B) Rs. 210
- (C) Rs. 216
- (D) Rs. 224
- (E) Rs. 230

Q3: What is the ratio of profit earned by the agency on one Heritage Tour package to one Adventure Tour package?

एक हेरिटेज टूर पैकेज और एक एडवेंचर टूर पैकेज पर एजेंसी द्वारा अर्जित लाभ का अनुपात क्या है?

- (A) 7 : 4
- (B) 4 : 9
- (C) 3 : 2
- (D) 2 : 1
- (E) 9 : 4

Q4: Find the sum of profit earned by the agency on one package of each of the given four tour packages.

दिए गए चारों टूर पैकेजों में से प्रत्येक के एक पैकेज पर एजेंसी द्वारा अर्जित लाभ का योग ज्ञात कीजिए।

- (A) Rs. 217
- (B) Rs. 206

- (C) Rs. 231
- (D) Rs. 224
- (E) Rs. 196

Q5: Profit earned by the agency on one Wildlife Tour package is what percent less than the profit earned on one Heritage Tour package?

एक वाइल्डलाइफ टूर पैकेज पर एजेंसी द्वारा अर्जित लाभ, एक हेरिटेज टूर पैकेज पर अर्जित लाभ से कितने प्रतिशत कम है?

- (A) 25%
- (B) $28 \frac{4}{7}\%$
- (C) $33 \frac{1}{3}\%$
- (D) 30%
- (E) 35%

Solutions:

Basic calculation to find profit per package**For Heritage Tour**

Total profit = $29400 \times 27\% = \text{Rs. } 7938$

Packages sold = 126

Profit per package = $7938 / 126 = \text{Rs. } 63$

For Wildlife Tour

Total profit = $29400 \times 21\% = \text{Rs. } 6174$

Packages sold = 147

Profit per package = $6174 / 147 = \text{Rs. } 42$

For Beach Tour

Total profit = $29400 \times 34\% = \text{Rs. } 9996$

Packages sold = 119

Profit per package = $9996 / 119 = \text{Rs. } 84$

For Adventure Tour

Total profit = $29400 \times 18\% = \text{Rs. } 5292$

Packages sold = 189

Profit per package = $5292 / 189 = \text{Rs. } 28$

Package	Profit %	Units Sold	Total Profit	Profit/Package
Heritage Tour	27%	126	Rs. 7938	Rs. 63
Wildlife Tour	21%	147	Rs. 6174	Rs. 42
Beach Tour	34%	119	Rs. 9996	Rs. 84
Adventure Tour	18%	189	Rs. 5292	Rs. 28

1. Option D

Total profit on Heritage Tour = $29400 \times 27\% = 7938$

Packages sold of Heritage Tour = 126

Profit per Heritage Tour package = $7938 \div 126 = 63$

Cost price per Heritage Tour package = 735

Selling price per Heritage Tour package = $735 + 63 = 798$

2. Option B

Total profit on Wildlife Tour = $29400 \times 21\% = 6174$

Profit per Wildlife Tour package = $6174 \div 147 = 42$

Total profit on Beach Tour = $29400 \times 34\% = 9996$

Profit per Beach Tour package = $9996 \div 119 = 84$

Let selling prices of Wildlife Tour and Beach Tour be $5x$ and $12x$ respectively

Cost price of Wildlife Tour = $5x - 42$

Cost price of Beach Tour = $12x - 84$

Given ratio = $2 : 5$

$(5x - 42) : (12x - 84) = 2 : 5$

$25x - 210 = 24x - 168$

$x = 42$

Selling price of one Wildlife Tour package = $5x = 210$

3. Option E

Profit per Heritage Tour package = 63

Total profit on Adventure Tour = $29400 \times 18\% = 5292$

Profit per Adventure Tour package = $5292 \div 189 = 28$

Required ratio = $63 : 28 = 9 : 4$

4. Option A

Profit per Heritage Tour package = 63

Profit per Wildlife Tour package = 42

Profit per Beach Tour package = 84

Profit per Adventure Tour package = 28

Required sum = $63 + 42 + 84 + 28 = 217$

5. Option C

Profit per Heritage Tour package = 63

Profit per Wildlife Tour package = 42

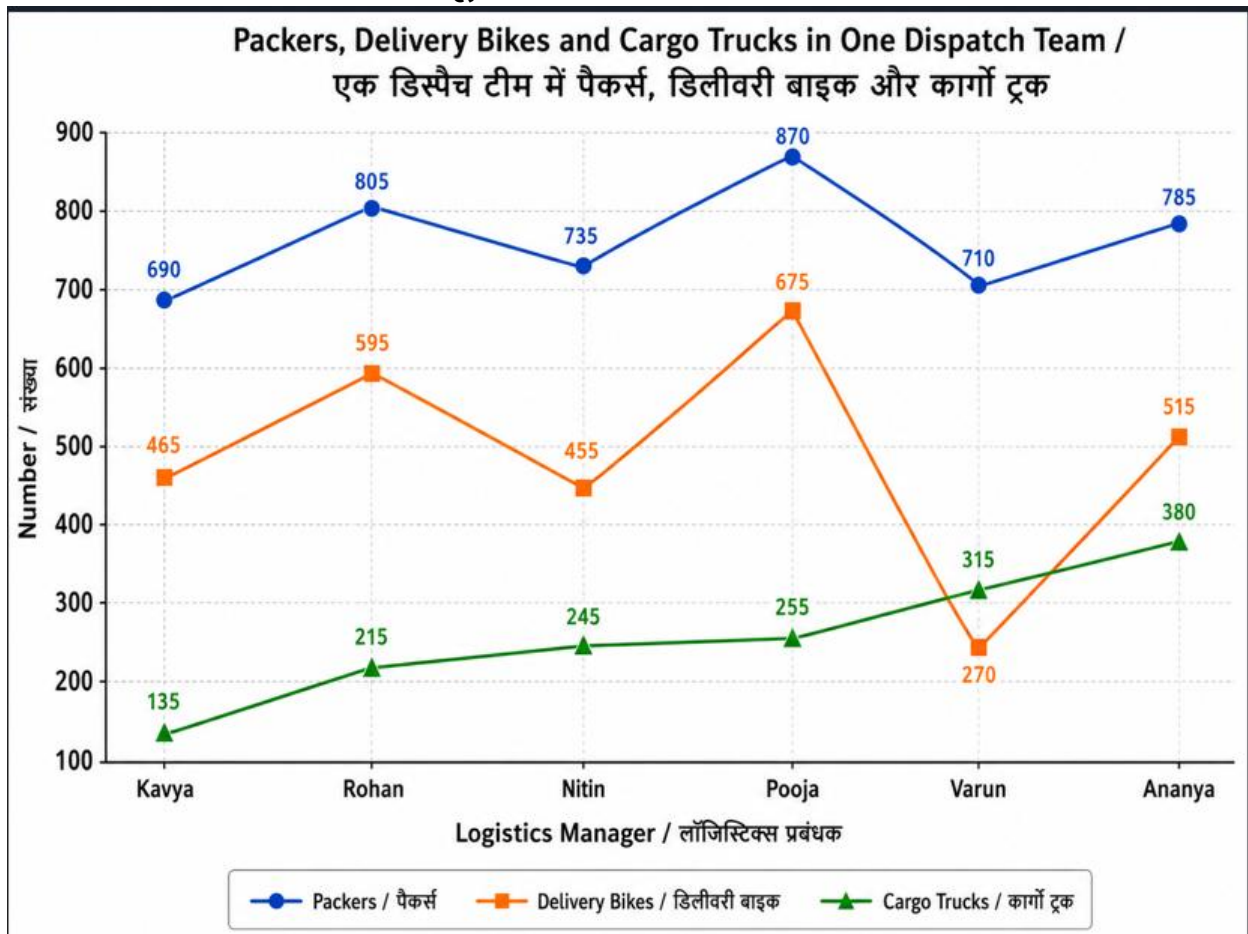
Difference = $63 - 42 = 21$

Required percentage = $21 \times 100 \div 63$

Required percentage = $33 \frac{1}{3}\%$

Set 3: The line chart shows the number of packers, delivery bikes and cargo trucks in one dispatch team of six logistics managers in a certain month. Each delivery bike is handled by 2 workers and each cargo truck is handled by 5 workers. Study the line chart data carefully and answer the questions.

रेखा आलेख एक निश्चित महीने में छह लॉजिस्टिक्स प्रबंधकों की एक डिस्पैच टीम में पैकर्स, डिलीवरी बाइक और कार्गो ट्रकों की संख्या दर्शाता है। प्रत्येक डिलीवरी बाइक को 2 कर्मचारी संभालते हैं और प्रत्येक कार्गो ट्रक को 5 कर्मचारी संभालते हैं। दिए गए रेखा आलेख के डेटा का ध्यानपूर्वक अध्ययन करें और प्रश्नों के उत्तर दें।



Q1: The ratio of the average number of workers in one dispatch team of Ananya to the average number of packers, delivery bikes and cargo trucks in one dispatch team of Rohan is $x:y$. Find $x + y$.

अनन्या की एक डिस्पैच टीम में कर्मचारियों की औसत संख्या और रोहन की एक डिस्पैच टीम में पैकर्स, डिलीवरी बाइक और कार्गो ट्रकों की औसत संख्या का अनुपात $x:y$ है। $x + y$ ज्ञात कीजिए।

(A) 1015

(B) 1090

- (C) 1037
- (D) 1066
- (E) 1112

Q2: Pooja gave one dispatch team to Kavya. Kavya sent that team along with one of her own dispatch teams to Varun. Varun distributed the packers, delivery bikes and cargo trucks of the two received teams equally among his 6 existing dispatch teams. Find the number of cargo trucks in each new dispatch team of Varun. All existing dispatch teams of a manager have the same data.

पूजा ने अपनी एक डिस्पैच टीम काव्या को दे दी। काव्या ने वह टीम अपनी एक डिस्पैच टीम के साथ वरुण को भेज दी। वरुण ने प्राप्त हुई दोनों टीमों के पैकर्स, डिलीवरी बाइक और कार्गो ट्रकों को अपनी 6 मौजूदा डिस्पैच टीमों में बराबर-बराबर बांट दिया। वरुण की प्रत्येक नई डिस्पैच टीम में कार्गो ट्रकों की संख्या ज्ञात कीजिए। किसी भी प्रबंधक की सभी मौजूदा डिस्पैच टीमों का डेटा समान है।

- (A) 380
- (B) 390
- (C) 405
- (D) 365
- (E) 395

Q3: By what percent is the total number of workers in one dispatch team of Kavya less than that of Varun?

काव्या की एक डिस्पैच टीम में कुल कर्मचारियों की संख्या वरुण की एक डिस्पैच टीम से कितने प्रतिशत कम है?

- (A) 15.76%
- (B) 16.76%
- (C) 17.76%
- (D) 18.76%
- (E) 19.76%

Q4: During a strike, $\frac{7}{10}$ of the total packers, delivery bikes and cargo trucks in one dispatch team of Ananya became inactive. If the ratio of the remaining active packers, delivery bikes and cargo trucks is 7:6:5, then find the number of active packers left.

हड़ताल के दौरान, अनन्या की एक डिस्पैच टीम के कुल पैकर्स, डिलीवरी बाइक और कार्गो ट्रकों में से 7/10 निष्क्रिय हो गए। यदि शेष सक्रिय पैकर्स, डिलीवरी बाइक और कार्गो ट्रकों का अनुपात 7:6:5 है, तो बचे हुए सक्रिय पैकर्स की संख्या ज्ञात कीजिए।

- (A) 182
- (B) 196
- (C) 210
- (D) 224
- (E) 238

Q5: In a large warehouse drill, one dispatch team of Nitin and one dispatch team of Pooja were arranged on two opposite sides of the central aisle. In each team, rows were formed such that each row contained the same maximum possible number of components of the same category, i.e. packers, delivery bikes or cargo trucks. Find the difference between the total number of rows formed in the two teams. The maximum possible number of components in each row is different for both teams.

एक बड़े गोदाम अभ्यास में, नितिन की एक डिस्पैच टीम और पूजा की एक डिस्पैच टीम को केंद्रीय गलियारे के दो विपरीत ओर व्यवस्थित किया गया। प्रत्येक टीम में पंक्तियां इस प्रकार बनाई गईं कि प्रत्येक पंक्ति में एक ही श्रेणी, अर्थात पैकर्स, डिलीवरी बाइक या कार्गो ट्रकों के समान अधिकतम संभव घटक हों। दोनों टीमों में बनी कुल पंक्तियों की संख्या का अंतर ज्ञात कीजिए। दोनों टीमों में प्रत्येक पंक्ति में रखे जाने वाले अधिकतम संभव घटकों की संख्या अलग-अलग है।

- (A) 72
- (B) 85
- (C) 79
- (D) 91
- (E) 66

Solutions:

1. Option D

$$\text{Ananya workers} = 785 + 2 \times 515 + 5 \times 380 = 3715$$

$$\text{Rohan total elements} = 805 + 595 + 215 = 1615$$

$$\text{Required ratio} = (3715/3):(1610/3) = 3715:1610 = 743:323$$

$$x + y = 743 + 323 = 1066$$

2. Option A

$$\text{Cargo trucks received by Varun} = 255 + 135 = 390$$

$$\text{Cargo trucks added to each of Varun's 6 teams} = 390/6 = 65$$

$$\text{New cargo trucks in each Varun team} = 315 + 65 = 380$$

3. Option D

$$\text{Kavya workers} = 690 + 2 \times 465 + 5 \times 135 = 2295$$

$$\text{Varun workers} = 710 + 2 \times 270 + 5 \times 315 = 2825$$

$$\text{Difference} = 2825 - 2295 = 530$$

$$\text{Required percentage} = 530/2825 \times 100 = 18.76\%$$

4. Option B

$$\text{Total elements in Ananya's team} = 785 + 515 + 380 = 1680$$

$$\text{Inactive elements} = 7/10, \text{ so active elements} = 3/10$$

$$\text{Active elements} = 1680 \times 3/10 = 504$$

$$\text{Ratio sum} = 7 + 6 + 5 = 18$$

$$\text{Active packers} = 504 \times 7/18 = 196$$

5. Option C

$$\text{Nitin maximum row size} = \text{HCF of } 735, 455 \text{ and } 245 = 35$$

$$\text{Rows in Nitin's team} = 735/35 + 455/35 + 245/35 = 21 + 13 + 7 = 41$$

$$\text{Pooja maximum row size} = \text{HCF of } 870, 675 \text{ and } 255 = 15$$

$$\text{Rows in Pooja's team} = 870/15 + 675/15 + 255/15 = 58 + 45 + 17 = 120$$

$$\text{Difference} = 120 - 41 = 79$$

Set 4: The data is about two types of products, Solar Lanterns and LED Bulbs, kept in three rural energy depots: Aravali Depot, Bhagirathi Depot and Charak Depot. Each depot has only these two types of products. The total stock in Aravali Depot is 390. The number of Solar Lanterns in Bhagirathi Depot is 13 more than the number of Solar Lanterns in Charak Depot. In Charak Depot, the number of Solar Lanterns is 104 less than the number of LED Bulbs. The number of LED Bulbs in Aravali Depot is 104 less than the number of Solar Lanterns in Bhagirathi Depot. The total stock in Charak Depot is 130 more than the total stock in Aravali Depot. The total number of LED Bulbs in all three depots together is 585.

यह डेटा तीन ग्रामीण ऊर्जा डिपो, अरावली डिपो, भागीरथी डिपो और चरक डिपो, में रखे गए दो प्रकार के उत्पादों, सौर लालटेन और LED बल्ब, से संबंधित है। प्रत्येक डिपो में केवल ये दो प्रकार के उत्पाद हैं। अरावली डिपो में कुल स्टॉक 390 है। भागीरथी डिपो में सौर लालटेन की संख्या, चरक डिपो में सौर लालटेन की संख्या से 13 अधिक है। चरक डिपो में सौर लालटेन की संख्या, LED बल्बों की संख्या से 104 कम है। अरावली डिपो में LED बल्बों की संख्या, भागीरथी डिपो में सौर लालटेन की संख्या से 104 कम है। चरक डिपो का कुल स्टॉक, अरावली डिपो के कुल स्टॉक से 130 अधिक है। तीनों डिपो में मिलाकर LED बल्बों की कुल संख्या 585 है।

Q1: If the simplified ratio of Solar Lanterns to LED Bulbs in Aravali Depot is $a : b$, find the value of $a + b$.

यदि अरावली डिपो में सौर लालटेन और LED बल्बों का सरलतम अनुपात $a : b$ है, तो $a + b$ का मान ज्ञात कीजिए।

- (A) 13
- (B) 8
- (C) 11
- (D) 10
- (E) 12

Q2: Find the difference between the total stock of Charak Depot and the total stock of Bhagirathi Depot.

चरक डिपो और भागीरथी डिपो के कुल स्टॉक के बीच अंतर ज्ञात कीजिए।

- (A) 143
- (B) 117
- (C) 156

- (D) 169
(E) 130

Q3: If the total number of Solar Lanterns in Aravali Depot and Bhagirathi Depot together is $p\%$ of the total stock in Charak Depot, find p .

यदि अरावली डिपो और भागीरथी डिपो में मिलाकर सौर लालटेन की कुल संख्या, चरक डिपो के कुल स्टॉक का $p\%$ है, तो p का मान ज्ञात कीजिए।

- (A) 90
(B) 95
(C) 85
(D) 100
(E) 105

Q4: If 39 LED Bulbs are transferred from Charak Depot to Aravali Depot, then what will be the total stock in Aravali Depot after the transfer?

यदि चरक डिपो से 39 LED बल्ब अरावली डिपो में स्थानांतरित कर दिए जाएँ, तो स्थानांतरण के बाद अरावली डिपो का कुल स्टॉक कितना होगा?

- (A) 390
(B) 416
(C) 403
(D) 442
(E) 429

Q5: If the total stock in Daksh Depot is the average of the total stock in Aravali Depot and Charak Depot, and Solar Lanterns in Charak Depot are 80% of Solar Lanterns in Daksh Depot, then find the number of LED Bulbs in Daksh Depot.

यदि दक्ष डिपो का कुल स्टॉक, अरावली डिपो और चरक डिपो के कुल स्टॉक का औसत है, और चरक डिपो में सौर लालटेन की संख्या, दक्ष डिपो में सौर लालटेन की संख्या का 80% है, तो दक्ष डिपो में LED बल्बों की संख्या ज्ञात कीजिए।

- (A) 182
(B) 208
(C) 195
(D) 169

(E) 221

Solutions:

Depot	Solar Lanterns	LED Bulbs	Total
Aravali Depot	273	117	390
Bhagirathi Depot	221	156	377
Charak Depot	208	312	520
Total	702	585	1287

1. Option D

Solar Lanterns in Aravali Depot = 273

LED Bulbs in Aravali Depot = 117

Required ratio = 273 : 117

= 7 : 3

 $a + b = 7 + 3$

= 10

2. Option A

Total stock in Charak Depot = 520

Total stock in Bhagirathi Depot = 377

Required difference = 520 – 377

= 143

3. Option B

Solar Lanterns in Aravali Depot and Bhagirathi Depot together = 273 + 221

= 494

Total stock in Charak Depot = 520

 $p = 494 \div 520 \times 100$

= 95

4. Option E

Initial total stock in Aravali Depot = 390

LED Bulbs transferred from Charak Depot to Aravali Depot = 39

Total stock in Aravali Depot after transfer = $390 + 39$
= 429

5. Option C

Total stock in Daksh Depot = $(390 + 520) \div 2$
= 455

Solar Lanterns in Charak Depot = 208

Solar Lanterns in Daksh Depot = $208 \div 80 \times 100$
= 260

LED Bulbs in Daksh Depot = $455 - 260$
= 195